

NAME: _____

PERIOD: _____

Which Pass Rate Answers Which Question?

AP Statistics · Unit 2 · Topic 2.2

[STAR] THE PROBLEM

Suppose a school records exam outcomes for 300 students. Students chose for themselves whether to use the tutoring center.

Mara: *The pass rate is 75%.*

Noah: *The pass rate is 88%.*

Elena: *Tutoring raises the pass rate by 13 percentage points because $88\% - 75\% = 13$ percentage points.*

Exam outcomes for 300 students

Center?	Passed	Did not pass	Total
Used	66	9	75
Did not	159	66	225
Total	225	75	300

FIRST TAKE — one sentence before you work the parts

Can 75% and 88% both be correct pass rates? State what group you think each percentage describes.

[BOOK] NAME THE DENOMINATOR, THEN THE CLAIM (keep on the page)

A **marginal relative frequency** uses a row or column total divided by the table total. A **conditional relative frequency** uses a cell count divided by the total for the named group. To decide whether two categorical variables are associated, compare the same response's conditional percentage across the groups. Different conditional distributions show association. They do not show cause when people choose their own groups; a fair causal comparison requires random assignment.

Work (a)–(c).

DOK 1 (a) Label 75 percent and 88 percent as marginal or conditional. Name the group for each.

DOK 2 (b) Verify the 75 percent and 88 percent rates. The nonuser pass rate is about 70.7 percent; find the user-minus-nonuser gap.

DOK 3 (c) Can Mara and Noah both be right? Explain their denominators. Use the two group rates to correct Elena’s 13-point comparison, and explain why her word *raises* goes beyond these data. Write 3–4 sentences.

Frame: Mara answers _____, while Noah answers _____.

Frame: The variables appear _____ because _____; the word *raises* is unwarranted because _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.